

NB45: Saglam Bayes-Error Ceiling (Cok Yontem + Floor-CI Disiplini)

SEED=42 | k-NN(coklu-k) + MST-GHP + Feature-Boot + Floor-CI | 2026-07-24

1. Amac ve Motivasyon

NB40, PAH panelinde tek yontemle (k-NN, tek k-seti) bir Bayes-error tavani hesaplamis ve gap=+0.1125 bulmustu -- bu, NB21-NB35'in 'PAH plato gercek' bulgusuyla celisiyordu. Literatur (arXiv 2506.03159), k-NN tabanlı Bayes-error tahmininin sinif basi ~100-1000 ornek gerektirdigini gosteriyor; PAH'in 62 benign ornegi bu esigin altinda. Bu notebook celiskiye (a) MST-tabanlı GHP ile capraz-dogrulama, (b) feature-alkume bootstrap ile kirligilicilik olcumu, (c) floor-F1'i ayni CI disipliniinde hesaplayarak cozer.

2. Yontemler-Arasi Uzlasma (k-NN vs MST-GHP)

Panel	kNN-1 R*_high	GHP Bayes-upper	Fark	Guvencilirlik
MASTER	0.2845	0.3824	0.0979	HIGH (n=782)
KANSER	0.1985	0.2413	0.0428	MEDIUM (n=120)
PAH	0.2124	0.3726	0.1602	LOW (n=62)
CFTR	0.1892	0.3671	0.1779	LOW (n=21)

3. Nihai Ozet: Floor vs Bayes vs Model (Ayni CI Disiplini)

Panel	Floor-F1[CI]	Bayes-F1[CI]	Model-F1	Gap(B-M)	Guvencilirlik Bayragi
MASTER	0.600 [0.600,0.600]	0.656 [0.648,0.668]	0.638	0.018	Bayes > Floor (anlamli sinyal var)
KANSER	0.562 [0.562,0.562]	0.726 [0.698,0.758]	0.730	-0.004	Bayes > Floor (anlamli sinyal var)
PAH	0.705 [0.705,0.705]	0.701 [0.685,0.715]	0.582	0.119	ARTEFAKT SUPHESI (Bayes~Floor, ayrisim yok)
CFTR	0.677 [0.677,0.677]	0.755 [0.727,0.772]	0.863	-0.108	Bayes > Floor (anlamli sinyal var)

4. PAH Ozel Karar

n_benign=62, reliability=LOW. Floor-F1=0.705 [0.7047619047619048, 0.7047619047619048], Bayes-F1=0.701 [0.6846278593376579, 0.714975845410628], Model-F1=0.582.

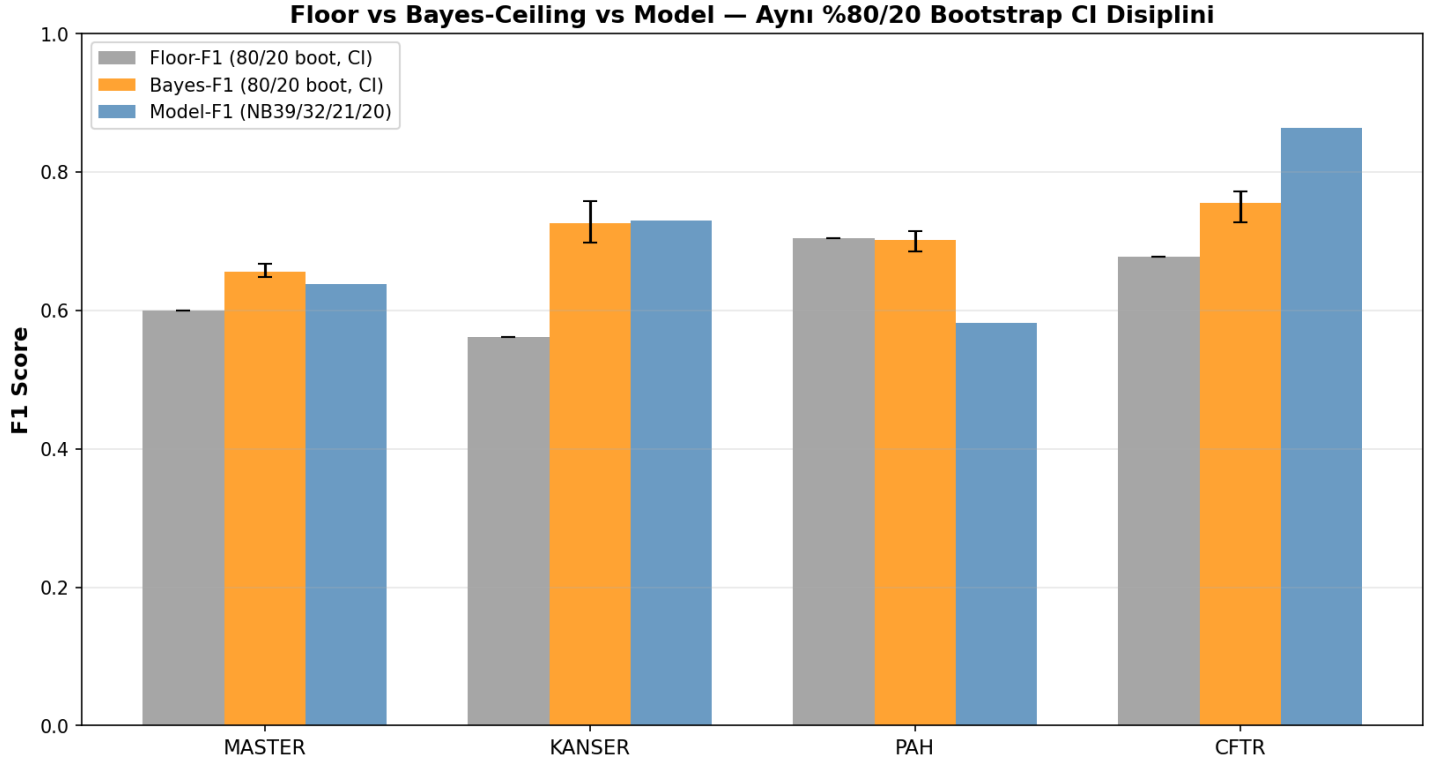
KARAR: ARTEFAKT: Bayes-F1 CI, Floor-F1 CI ile cakisiyor (veya guvenilirlik LOW). NB40'in gap=+0.1125 bulgusu k-NN artefaktidir. PAH plato bulgusu (NB21-NB35, danisman floor-analizi) GECERLIGINI KORUR.

NB45: Saglam Bayes-Error Ceiling (Cok Yontem + Floor-CI Disiplini)

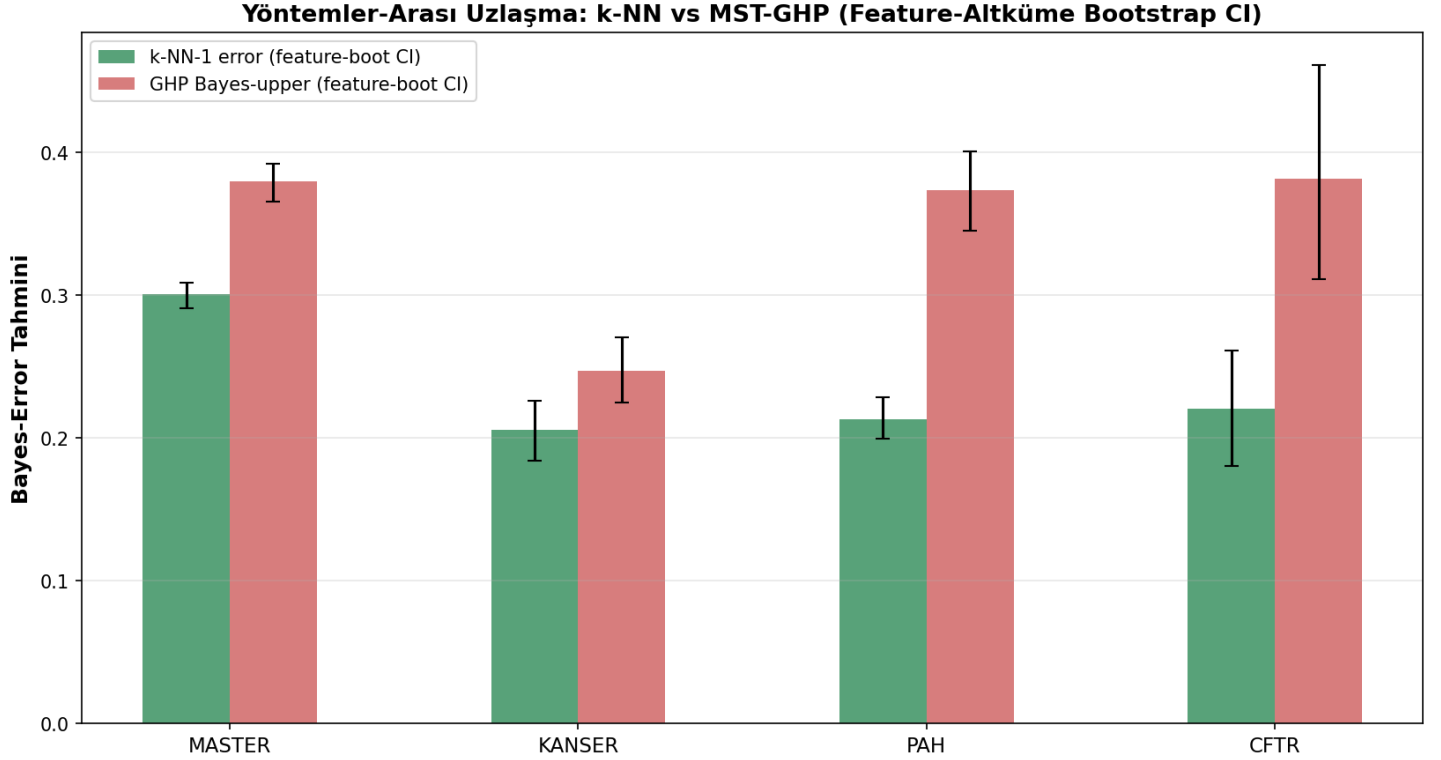
SEED=42 | k-NN(coklu-k) + MST-GHP + Feature-Boot + Floor-CI | 2026-07-24

5. Gorsellestirilmis Sonuclar

Sekil 1: fig1_floor_bayes_model.png



Sekil 2: fig2_method_convergence.png

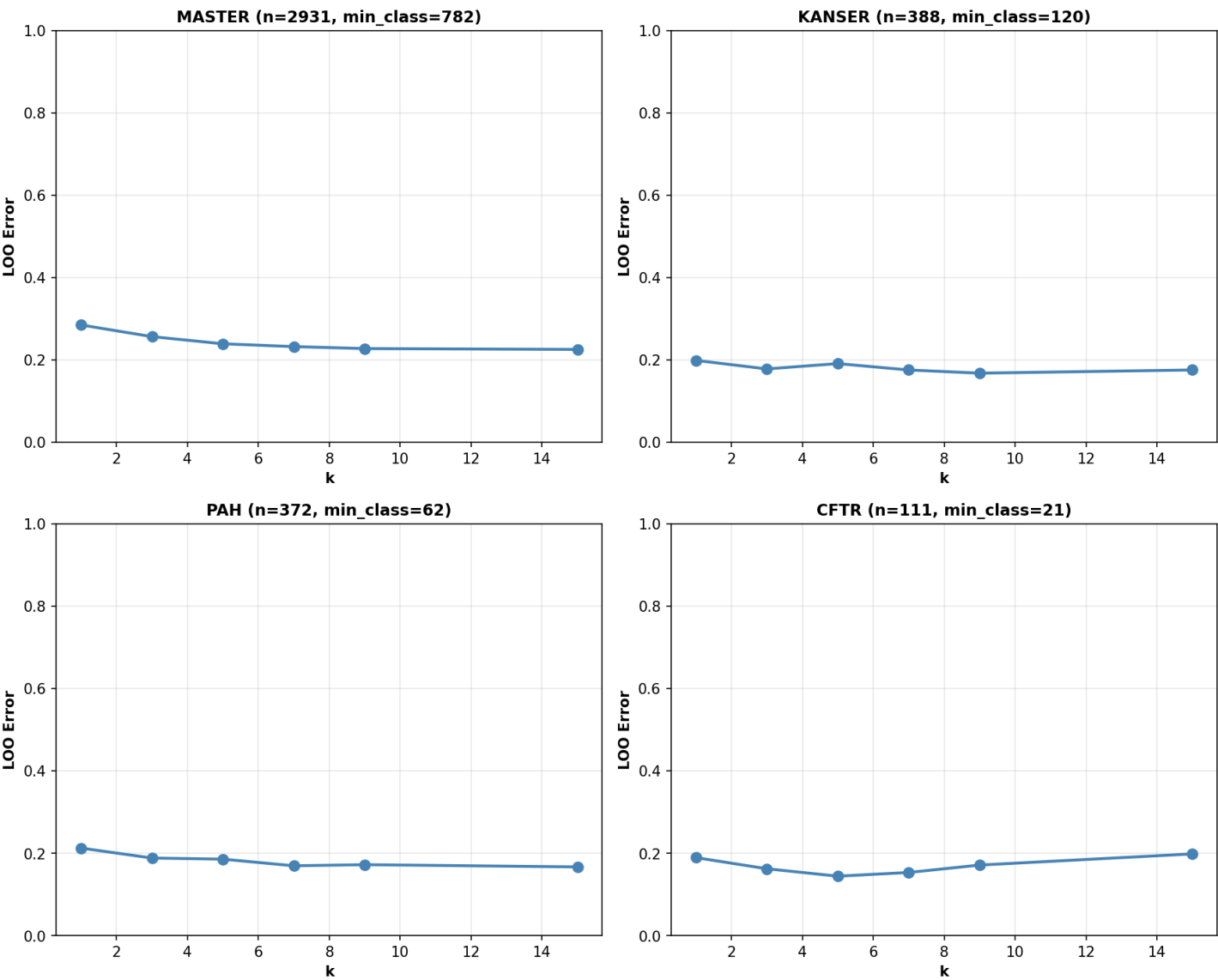


Sekil 3: fig3_knn_curve_extended.png

NB45: Saglam Bayes-Error Ceiling (Cok Yontem + Floor-CI Disiplini)

SEED=42 | k-NN(coklu-k) + MST-GHP + Feature-Boot + Floor-CI | 2026-07-24

k-NN LOO Hata Egrisi (k=1..15)



Sekil 4: fig4_gap_vs_floor_artefact_test.png

NB45: Sağlam Bayes-Error Ceiling (Cok Yontem + Floor-CI Disiplini)

SEED=42 | k-NN(coklu-k) + MST-GHP + Feature-Boot + Floor-CI | 2026-07-24

